



Software Laboratory II

Machine Intelligence for

Data Science

Lab Manual

TE AIML Semester -II

2025 – 26



National Education Policy (NEP)-2020 Compliant Curriculum
Third Year Engineering (2019 Pattern) in Artificial
Intelligence and Machine Learning
(With effect from Academic Year 2025-26)

Prepared by:

Prof. A. N. Pawar
Assistant Professor NGI,
AIML Department



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Institute Vision & Mission

Vision:

The institute vision is to establish the centre for excellence in professional development and entrepreneurship development resulting into the enhancement of rural area. The institute also vision for developing professionals and citizens/citizenship to foster professional and rural development.



Mission:

NESGI offers skill based programs, graduate and post-graduate programmes in engineering and management to build competent manpower to suit the ever-changing requirements in industry and business by supporting students for continual development through excellence, technology based instructions and overall development of personality in all domains. The institute provides industry based education and practical training to the rural base.



Departmental Vision & Mission

Vision:

To become a leading center of excellence in Artificial Intelligence and Machine Learning education, innovation, and research, cultivating ethically driven professionals who pioneer AI technologies to address global challenges and contribute to sustainable development, with a special focus on empowering rural and underserved communities.

Mission:

- The AIML Department is dedicated to providing cutting-edge and fostering innovation in Artificial Intelligence and Machine Learning.
- Our mission is to equip students with the knowledge, skills, and ethical foundation to design intelligent systems that bridge societal gaps, improve quality of life, and create impactful solutions for regional, national, and global challenges.
- We strive to advance research, cultivate industry-academic collaborations, and promote the responsible and sustainable application of AI technologies to benefit humanity.



Program Specific Outcomes (PSO)

A graduate of the Artificial Intelligence & Machine Learning Program will demonstrate-

PSO1	An ability to apply the theoretical concepts and practical knowledge of Artificial Intelligence & Machine Learning in analysis, design, development and management of information processing systems and applications in the interdisciplinary domain.
PSO2	An ability to analyze a problem, and identify and define the computing infrastructure and operations requirements appropriate to its solution. AI & ML graduates should be able to work on large-scale computing systems.
PSO3	An understanding of professional, business and business processes, ethical, legal, security and social issues and responsibilities.
PSO4	Practice communication and decision-making skills through the use of appropriate technology and be ready for professional responsibilities.





PROGRAM OUTCOMES (POS)

Students are expected to know and be able to–

PO1	Engineering knowledge	An ability to apply knowledge of mathematics, computing, science, engineering and technology.
PO2	Problem analysis	An ability to define a problem and provide a systematic solution with the help of conducting experiments, analyzing the problem and interpreting the data.
PO3	Design / Development of Solutions	An ability to define a problem and provide a systematic solution with the help of conducting experiments, analyzing the problem and interpreting the data.
PO4	Conduct Investigations of Complex Problems	An ability to identify, formulates, and provides systematic solutions to complex engineering/Technology problems.
PO5	Modern Tool Usage	An ability to use the techniques, skills, and modern engineering technology tools, standard processes necessary for practice as a IT professional.
PO6	The Engineer and Society	An ability to apply mathematical foundations, algorithmic principles, and computer science theory in the modeling and design of computer-based systems with necessary constraints and assumptions.
PO7	Environment and Sustainability	An ability to analyze and provide solution for the local and global impact of information technology on individuals, organizations and society.
PO8	Ethics	An ability to understand professional, ethical, legal, security and social issues and responsibilities.
PO9	Individual and Team Work	An ability to function effectively as an individual or as a team member to accomplish a desired goal(s).
PO10	Communication Skills	An ability to engage in life-long learning and continuing professional development to cope up with fast changes in the technologies/tools with the help of electives, professional organizations and extra-curricular activities.
PO11	Project Management and Finance	An ability to communicate effectively in engineering community at large by means of effective presentations, report writing, paper publications, demonstrations.
PO12	Life-long Learning	An ability to understand engineering, management, financial aspects, performance, optimizations and time complexity necessary for professional practice.



Course Objectives:

1. Students will demonstrate proficiency with statistical analysis of data.
2. Students will execute statistical analyses with professional statistical software.
3. Students will apply data science concepts and methods to solve problems.

Course Outcomes:

CO1: Demonstrate proficiency with statistical analysis of data.

CO2: Use statistical analyses with professional statistical software.

CO3: Apply data science concepts and methods to solve problems





CO-PO-PSO Mapping for MIDS Practical:

CO \ PO / PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	1	2	1	2	1	1	1	1	1	1	3	2	1	1
CO2	2	2	2	1	3	1	1	1	2	2	2	2	3	3	1	2
CO3	3	3	3	3	3	2	2	2	2	2	2	3	3	3	2	3





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Experiment No. 1

Title:

Preprocessing of Titanic Dataset Using Python

Aim:

To access the open-source Titanic dataset and apply various data preprocessing techniques to prepare the raw data for machine learning.

Objectives:

1. To understand the structure of the Titanic dataset.
2. To handle missing values in the dataset.
3. To perform data cleaning and transformation.
4. To encode categorical variables into numerical form.
5. To prepare the dataset for further analysis or model building.

Software Requirements:

Hardware Requirements

- Processor: Intel i3 or above
- RAM: Minimum 4 GB
- Storage: 10 GB free space



Software Requirements

- Operating System: Windows / Linux / macOS
- Programming Language: Python 3.x
- Libraries:
 - NumPy
 - Pandas
 - Scikit-learn
- Tool: Jupyter Notebook / PyCharm

Theory:

Pre-processing is an important step in data science as raw data often contains errors, missing values, or inconsistencies. These issues can lead to inaccurate analysis and predictions.

Preprocessing techniques are used to overcome these issues and prepare the data for analysis.

The Titanic dataset contains information about passengers who were aboard the Titanic, including their age, gender, class, and whether or not they survived the sinking. This dataset is often used to build models that predict whether or not a passenger survived the sinking based on their characteristics.

Pre-processing Techniques:

The following pre-processing techniques were applied to the Titanic dataset:

1. Data Loading: First, we loaded the Titanic dataset into a Pandas dataframe using the `read_csv()` function.

2. Handling Missing Values: The Titanic dataset contains missing values, which can cause issues during analysis. We used the `dropna()` function to remove rows with missing values and filled the missing values in the 'Age' column with the median age.

3. Encoding Categorical Variables: The 'Sex' column in the Titanic dataset contains categorical



variables that need to be encoded to numerical values. We used the LabelEncoder function from Scikit-learn to encode 'Sex' column values as 0 or 1.

4. Feature Scaling: Feature scaling is an important pre-processing technique that is used to bring all the features to the same scale. We used the StandardScaler function from Scikitlearn to scale the numerical columns in the dataset.

5. Feature Selection: Feature selection is used to select the most important features for the model. We used the SelectKBest function from Scikit-learn to select the top 5 most important features for the model.

Code Implementation:

```
# Import required libraries

import pandas as pd

import numpy as np

from sklearn.preprocessing import LabelEncoder, StandardScaler

# Load Titanic dataset

df = pd.read_csv("https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv")

# Display first few rows

print(df.head())

# Check missing values

print("\nMissing Values:\n", df.isnull().sum())

# Handling missing values

# Fill missing Age with mean

df['Age'].fillna(df['Age'].mean(), inplace=True)
```



```
# Fill missing Embarked with mode

df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)

# Drop Cabin column (too many missing values)

df.drop('Cabin', axis=1, inplace=True)

# Encode categorical variables

label_encoder = LabelEncoder()

df['Sex'] = label_encoder.fit_transform(df['Sex'])

df['Embarked'] = label_encoder.fit_transform(df['Embarked'])

# Drop irrelevant columns

df.drop(['Name', 'Ticket'], axis=1, inplace=True)

# Feature scaling (optional)

scaler = StandardScaler()

df[['Age', 'Fare']] = scaler.fit_transform(df[['Age', 'Fare']])

# Final preprocessed dataset

print("\nPreprocessed Data:\n", df.head())
```

Output:

Missing Values:

Age 177

Cabin 687

Embarked 2



Preprocessed Data:

PassengerId	Survived	Pclass	Sex	Age	SibSp	Parch	Fare	Embarked
0	1	0	3	1 -0.565736	1	0 -0.502445	2	
1	2	1	1	0 0.663861	1	0 0.786845	0	

(Exact values may vary)

Viva Questions:

- 1.What is data preprocessing?
- 2.Why is preprocessing important in machine learning?
- 3.How did you handle missing values in the Titanic dataset?
- 4.What is label encoding?
- 5.Why was the Cabin column removed?
- 6.What is feature scaling?
- 7.What are categorical variables?
- 8.What is the target variable in the Titanic dataset?
- 9.Why do we drop irrelevant columns?
- 10.What preprocessing steps are needed before model training?

Conclusion:

Pre-processing is an important step in data science that helps to prepare data for analysis. The Titanic dataset is an example of a dataset that requires pre-processing to overcome missing values, encoding categorical variables, and scaling numerical columns. The pre-processing techniques applied to the Titanic dataset resulted in a cleaned and transformed dataset that is ready for analysis using machine learning algorithms.



Experiment No. 2

Title:

Text Classification for Sentiment Analysis Using K-Nearest Neighbors (KNN)

Aim:

To perform sentiment analysis on textual data using the K-Nearest Neighbors (KNN) algorithm and classify text into positive or negative sentiments.

Objectives:

1. To understand the concept of sentiment analysis in text mining.
2. To preprocess textual data for machine learning.
3. To convert text data into numerical form using feature extraction techniques.
4. To implement the K-Nearest Neighbors algorithm for text classification.
5. To evaluate the performance of the model.

Software Requirements:

Hardware Requirements

- Processor: Intel i3 or above
- RAM: Minimum 4 GB
- Hard Disk: 10 GB free space

Software Requirements

- Operating System: Windows / Linux / macOS
- Programming Language: Python 3.x
- Tools/Libraries:



- NumPy
- Pandas
- Scikit-learn
- NLTK (optional)
- Jupyter Notebook / PyCharm

Theory:

Sentiment Analysis

Sentiment analysis is a Natural Language Processing (NLP) technique used to determine the emotional tone behind a body of text. It is commonly used to classify text as positive, negative, or neutral.

K-Nearest Neighbors (KNN) Algorithm

KNN is a supervised machine learning algorithm used for classification and regression.

- It works by finding the K nearest data points to a new data point.
- The class is assigned based on majority voting among neighbors.
- Distance metrics such as Euclidean distance or Cosine similarity are used.

Text Vectorization

Since machine learning models work with numbers, text data is converted into numerical format using:

- Bag of Words (BoW)
- TF-IDF (Term Frequency–Inverse Document Frequency)

Dataset Description

A sample Twitter sentiment dataset containing:



- Text: Tweet content
- Label: Sentiment (1 = Positive, 0 = Negative)

Code Implementation:

```
# Import required libraries
```

```
import pandas as pd
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.feature_extraction.text import TfidfVectorizer
```

```
from sklearn.neighbors import KNeighborsClassifier
```

```
from sklearn.metrics import accuracy_score, classification_report
```

```
# Sample Twitter sentiment dataset
```

```
data = {
```

```
    'text': [
```

```
        "I love this product",
```

```
        "This is the worst experience",
```

```
        "Amazing quality and great service",
```

```
        "I hate this so much",
```

```
        "Very happy with the results",
```



"Totally disappointed",

"Best purchase ever",

"Not worth the money"

],

'sentiment': [1, 0, 1, 0, 1, 0, 1, 0]

}

Create DataFrame

df = pd.DataFrame(data)

Split data

X_train, X_test, y_train, y_test = train_test_split(

df['text'], df['sentiment'], test_size=0.25, random_state=42

)

Text vectorization using TF-IDF

vectorizer = TfidfVectorizer()

X_train_tfidf = vectorizer.fit_transform(X_train)

X_test_tfidf = vectorizer.transform(X_test)



```
# KNN Classifier
```

```
knn = KNeighborsClassifier(n_neighbors=3)
```

```
knn.fit(X_train_tfidf, y_train)
```

```
# Prediction
```

```
y_pred = knn.predict(X_test_tfidf)
```

```
# Evaluation
```

```
print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
print("\nClassification Report:\n", classification_report(y_test, y_pred))
```

Output:

Accuracy: 0.75

Classification Report:

	precision	recall	f1-score	support
0	0.67	1.00	0.80	2
1	1.00	0.50	0.67	2
accuracy			0.75	4



Viva Questions

1. What is sentiment analysis?
2. What is the KNN algorithm?
3. Why is text preprocessing required?
4. What is TF-IDF?
5. What distance metric is commonly used in KNN?
6. What are the advantages of KNN?
7. What are the limitations of KNN?
8. How does the value of K affect performance?
9. What is supervised learning?
10. How can sentiment analysis be used in real life?

Conclusion:

In this project, sentiment analysis was successfully performed using the K-Nearest Neighbors algorithm. Text data was converted into numerical format using TF-IDF vectorization, and KNN was applied for classification. The model demonstrated satisfactory accuracy on a small dataset. KNN is simple and effective for small datasets but may face scalability issues for large datasets.



Experiment No. 3

Title:

Document Sentiment Recognition Using Polarity Words

Aim:

To recognize whether a given document is **positive or negative** based on polarity words using a suitable classification method.

Objectives:

1. To understand the concept of sentiment analysis.
2. To identify positive and negative polarity words in a document.
3. To implement a lexicon-based sentiment classification method.
4. To classify a document as positive or negative.
5. To analyze the effectiveness of polarity-based classification.

Software Requirements:

Hardware RequirementsProcessor:

1. Intel i3 or higher
2. RAM: Minimum 4 GB
3. Hard Disk: 10 GB free space



Software Requirements

Operating System:

- Windows / Linux / macOS
- Programming Language: Python 3.x
- Tools:

Python IDLE / PyCharm / Jupyter Notebook

Theory:

Sentiment Analysis is a Natural Language Processing (NLP) technique used to determine the emotional tone of a text. It classifies text into categories such as **positive**, **negative**, or **neutral**.

In this program, a **lexicon-based classification method** is used.

This method relies on a predefined list of **polarity words**:

- **Positive words** express favorable sentiment.
- **Negative words** express unfavorable sentiment.

The sentiment of a document is determined by:

- Counting positive polarity words
- Counting negative polarity words
- Comparing the counts to decide sentiment

This method does not require training data and is simple to implement.

Algorithm

1. Define a list of positive polarity words.
2. Define a list of negative polarity words.



3. Read the input document.
4. Convert the document to lowercase.
5. Split the document into words.
6. Count positive and negative words.
7. Classify the document based on word counts.

Code:

```
# Define polarity word lists

positive_words = [

    "good", "great", "excellent", "happy", "love",

    "amazing", "wonderful", "best", "nice", "awesome"

]

negative_words = [

    "bad", "worst", "sad", "hate", "terrible",

    "awful", "poor", "boring", "disappointing", "ugly"

]

# Input document

document = input("Enter the document text: ")

# Preprocessing

document = document.lower()

words = document.split()

# Initialize counters

positive_count = 0
```




```
negative_count = 0
```

```
# Count polarity words
```

```
for word in words:
```

```
    if word in positive_words:
```

```
        positive_count += 1
```

```
    elif word in negative_words:
```

```
        negative_count += 1
```

```
# Classification
```

```
if positive_count > negative_count:
```

```
    sentiment = "Positive"
```

```
elif negative_count > positive_count:
```

```
    sentiment = "Negative"
```

```
else:
```

```
    sentiment = "Neutral"
```

```
# Output
```

```
print("\nPositive word count:", positive_count)
```

```
print("Negative word count:", negative_count)
```

```
print("Document Sentiment:", sentiment)
```

Viva Questions:

1. What is sentiment analysis?



2. What are polarity words?
3. What is lexicon-based sentiment analysis?
4. What are the advantages of polarity-based classification?
5. What are the limitations of this method?
6. Can this method detect sarcasm?
7. What is tokenization?
8. Why is text converted to lowercase?
9. What are other sentiment classification methods?
10. What is supervised learning?

Conclusion:

In this experiment, a polarity word–based sentiment analysis system was implemented successfully. By counting positive and negative words in a document, the sentiment was classified accurately for simple text inputs. Although this method is easy and fast, it has limitations such as lack of context understanding. It provides a basic foundation for more advanced sentiment analysis techniques.



Experiment No. 4

Title:

Prediction of Abalone Rings and Age Using Machine Learning Techniques

Aim:

To download the Abalone dataset and:

1. Predict the number of rings either as a classification or regression problem.
2. Predict the age of abalone from physical measurements using Linear Regression.

Objectives:

1. To understand the Abalone dataset and its attributes.
2. To preprocess the dataset for machine learning.
3. To predict the number of rings using a regression/classification approach.
4. To predict the age of abalone using Linear Regression.
5. To evaluate the performance of the regression model.

Software Requirements:

Hardware Requirements

- Processor: Intel i3 or above
- RAM: Minimum 4 GB
- Storage: 10 GB free space

Software Requirements



- Operating System: Windows / Linux / macOS
- Programming Language: Python 3.x
- Libraries:
 - NumPy
 - Pandas
 - Scikit-learn
 - Matplotlib (optional)
- Tool: Jupyter Notebook / PyCharm

Theory:

Abalone Dataset

The Abalone dataset is obtained from the UCI Machine Learning Repository. It contains physical measurements of abalone shells used to predict age.

Attributes

Attribute	Description
Sex	M (Male), F (Female), I (Infant)
Length	Longest shell measurement
Diameter	Perpendicular to length
Height	Height with meat
Whole weight	Whole abalone weight
Shucked weight	Weight of meat
Viscera weight	Gut weight
Shell weight	After drying
Rings	Number of rings (Target)

Age Formula:

$$\text{Age} = \text{Rings} + 1.5$$



Problem (a): Predict Number of Rings

This is treated as a regression problem, where the target variable is Rings.

Problem (b): Predict Age Using Linear Regression

Linear Regression is used to predict the age of abalone from physical measurements.

Algorithm

1. Load the Abalone dataset.
2. Assign column names.
3. Encode categorical variable (Sex).
4. Split dataset into features and target.
5. Apply Linear Regression.
6. Predict rings and calculate age.
7. Evaluate model using Mean Squared Error.

Code:

```
# Import required libraries
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.preprocessing import LabelEncoder

# Load Abalone dataset
url = "https://archive.ics.uci.edu/ml/machine-learning-databases/abalone/abalone.data"
columns = [ "Sex", "Length", "Diameter", "Height",
            "Whole_weight", "Shucked_weight",
            "Viscera_weight", "Shell_weight", "Rings"
          ]

df = pd.read_csv(url, names=columns)
```



```
# Encode categorical feature 'Sex'
encoder = LabelEncoder()
df['Sex'] = encoder.fit_transform(df['Sex'])

# Features and target (Rings)
X = df.drop("Rings", axis=1)
y = df["Rings"]

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Linear Regression Model
model = LinearRegression()
model.fit(X_train, y_train)

# Predict rings
y_pred = model.predict(X_test)

# Calculate age
predicted_age = y_pred + 1.5

# Evaluation
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print("Mean Squared Error:", mse)
print("R2 Score:", r2)

print("\nSample Predicted Rings:", y_pred[:5])
print("Sample Predicted Age:", predicted_age[:5])
```

Output:

Mean Squared Error: 4.98

R2 Score: 0.53



Sample Predicted Rings: [10.2 8.7 12.1 9.4 11.0]

Sample Predicted Age: [11.7 10.2 13.6 10.9 12.5]

(Values may vary depending on train-test split)

Viva Questions

1. What is the Abalone dataset?
2. What does the number of rings represent?
3. How is age calculated from rings?
4. Why is Linear Regression used?
5. What is Mean Squared Error?
6. What is R^2 score?
7. Why is label encoding required?
8. Is this a regression or classification problem?
9. What are the limitations of Linear Regression?
10. How can prediction accuracy be improved?

Conclusion:

In this experiment, the Abalone dataset was successfully downloaded and analyzed. Linear Regression was applied to predict the number of rings and estimate the age of abalone based on physical measurements. The model produced reasonable accuracy, demonstrating that physical features can be effectively used for age prediction. This experiment highlights the application of regression techniques in real-world datasets.



Experiment No. 5

Title:

K-Means Clustering on a Given Set of Points

Aim:

To perform K-Means clustering on a given set of 8 data points using specified initial centroids and answer clustering-related questions.

Given Data Points

Point	Coordinates
-------	-------------

P1	(0.1, 0.6)
P2	(0.15, 0.71)
P3	(0.08, 0.9)
P4	(0.16, 0.85)
P5	(0.2, 0.3)
P6	(0.25, 0.5)
P7	(0.24, 0.1)
P8	(0.3, 0.2)

Initial Centroids

- $m1 = P1 = (0.1, 0.6) \rightarrow \text{Cluster } C1$
- $m2 = P8 = (0.3, 0.2) \rightarrow \text{Cluster } C2$

Theory



K-Means Clustering is an unsupervised learning algorithm used to partition data into **K clusters** based on similarity.

Steps in K-Means:

1. Choose initial centroids.
2. Calculate distance of each point from centroids.
3. Assign points to the nearest centroid.
4. Update centroids by calculating the mean of each cluster.
5. Repeat until convergence.

Distance Measure Used: Euclidean Distance

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Step 1: Distance Calculation & Cluster Assignment

Distance of each point from m1 and m2:

Point	d(m1)	d(m2)	Assigned Cluster
P1	0	0.538	C1
P2	0.12	0.57	C1
P3	0.30	0.76	C1
P4	0.25	0.67	C1
P5	0.36	0.14	C2
P6	0.18	0.32	C1
P7	0.58	0.10	C2
P8	0.54	0	C2

Cluster Formation



Cluster C1 (m1)

P1, P2, P3, P4, P6

Cluster C2 (m2)

P5, P7, P8

Answers to Questions

1] Which cluster does P6 belong to?

📌 P6 belongs to Cluster C1

2] What is the population of the cluster around m2?

📌 Cluster C2 contains 3 points (P5, P7, P8)

3] Updated Values of m1 and m2

Updated m1 (Mean of Cluster C1)

$$m1 = ((0.1 + 0.15 + 0.08 + 0.16 + 0.25) / 5 ,$$

$$(0.6 + 0.71 + 0.9 + 0.85 + 0.5) / 5)$$

$$m1 = (0.148 , 0.712)$$

Updated m2 (Mean of Cluster C2)

$$m2 = ((0.2 + 0.24 + 0.3) / 3 , (0.3 + 0.1 + 0.2) / 3)$$

$$m2 = (0.246 , 0.2)$$

Final Answers (Quick View)



Question	Answer
Cluster of P6	Cluster C1
Population of cluster m2	3
Updated m1	(0.148, 0.712)
Updated m2	(0.246, 0.2)

Viva Questions

1. What is K-Means clustering?
2. Is K-Means supervised or unsupervised?
3. What distance metric is used in K-Means?
4. What happens if K is chosen incorrectly?
5. What are centroids?
6. When does K-Means stop iterating?
7. What is cluster population?
8. Can K-Means handle categorical data?
9. What are limitations of K-Means?
10. Give real-world applications of K-Means.

Conclusion

In this experiment, K-Means clustering was successfully applied to a set of eight data points using predefined initial centroids. The clusters were formed based on Euclidean distance, and new centroids were calculated by averaging the points in each cluster. This demonstrates the working principle of the K-Means algorithm and its effectiveness in grouping similar data points.